

RESEARCH

Open Access



Global market trends and financial performance of the corporate fast-food industry and their potential contributions to diets high in meat and ultra-processed foods

Katherine Sievert^{1*}, Tanita Botha², Benjamin Wood¹, Phillip Baker³, Tailane Scapin¹ and Gary Sacks¹

Abstract

Background The fast-food industry has transformed substantially in recent decades – from diverse, locally rooted providers into a globalised, and increasingly corporate-led industry. Corporate fast-food retailers (FFRs) represent a key retail channel through which both ultra-processed foods and intensively produced animal source foods are consumed and normalised within corporate-industrial food systems. These dietary patterns are strongly associated with increased risks of diet-related diseases and contribute significantly to environmental degradation, including greenhouse gas emissions, land use change, and biodiversity loss. Despite the growing significance of FFRs, there has been limited analysis of their financial strategies and implications for global food system transformation.

Results We conducted a global analysis of market data from 54 countries and financial data of publicly listed FFRs, examining trends in FFR sales (2009–2023), market dominance, and the financial performance of leading publicly listed firms (1980–2023). We found that while sales in high-income countries were stagnating, leading firms maintained stable net profit margins and delivered relatively high shareholder returns, facilitated by financial strategies such as franchising and private equity ownership. U.S.-based corporations dominated the global market, with substantial expansion into countries outside the global North. These trends reflect the consolidation of power within the corporate food regime.

Conclusions The global expansion of corporate FFRs underscore their growing influence over diets and food systems, with critical implications for public health, ecological sustainability, and social justice. Policies targeting structural leverage points, for example, democratising corporate governance, reducing the influence of private equity, and re-orienting agri-food subsidies, are essential to countering the entrenchment of this model and supporting more democratic and sustainable food systems.

Clinical trial number Not applicable.

*Correspondence:
Katherine Sievert
k.sievert@deakin.edu.au

Full list of author information is available at the end of the article



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Keywords Fast food, Meat consumption, Ultra-processed food, Sustainable food systems, Corporate consolidation

Introduction

Healthy and sustainable food systems are those “that deliver food security and nutrition for all in such a way that the economic, social and environmental bases to generate food security and nutrition for future generations are not compromised” [1]. Diets arising from such systems are diverse but generally characterised by an adequate intake of whole and minimally processed foods such as fruits, vegetables, legumes, whole grains, and some animal source foods, while limiting foods that are harmful to health or produced in ways that degrade ecosystems, and exploit workers and non-human animals [2].

In contrast, corporate-industrial food systems, which are dominant in many countries, are recognised drivers of unhealthy and unsustainable diets, including high amounts of intensively produced animal-source food (ASF) and ultra-processed food¹ (UPF). These systems, shaped by increasing financialisation and corporate consolidation² of food production, distribution, and retail, prioritise profit maximisation, while externalising much of the true costs of intensive ASF and UPF production onto society and the natural environment [3, 4]. Understanding the corporative imperatives and financial logics underpinning the proliferation of these foods is an important aspect of food systems research, including examination of how corporate food actors and their practices shape both dietary patterns and broader food system transformations. This focus is critical given that population diets that are high in meat and UPFs (including ultra-processed beverages and meat) are linked with intersecting public health crises, including the rising burden of non-communicable diseases, antimicrobial resistance, and novel zoonotic diseases [5–7].

Inputs required to mass produce both categories of products place undue strain on the environment, including excessive greenhouse gas emissions, deforestation, biodiversity loss, and pollution [8–10]. Monoculture farming – and intensive farming more broadly – is associated with adverse effects on the welfare of farmers, workers and non-human animals involved in production, in addition to the displacement and endangerment of native wildlife [11–13]. These social and environmental externalities are increasingly understood not as incidental, but as systemic features of contemporary

corporate-industrial food systems [14]. Thus, tackling the global dominance of corporate-industrial food systems, and reducing production and consumption of UPFs and intensively produced ASFs, has been proposed as a critical component to realising healthy, sustainable and just food systems [15, 16]. While ASFs and UPFs are increasingly a focus of international policy agendas, including the World Health Organization (WHO) and the Food and Agriculture Organization of the United Nations (FAO) [7, 17–19], consensus on the role of meat in healthy and sustainable diets remains contested, particularly in relation to nutritional benefits and environmental impacts. It is broadly acknowledged that ASFs can play an important role in nutrition and that certain production methods can be sustainable [20].

Within corporate-industrial food systems, fast food, which is typically characterised by both UPFs and intensively produced ASFs, represents a key retail channel through which these products are consumed and normalised [21–24]. Dixon (2024) highlights the McDonald’s ‘McNugget’ as emblematic of how corporate fast-food retailers (FFRs) have transformed raw agricultural commodities into convenient and ultra-processed offerings, tailored to the fast-food model [25]. While small-scale and local food vendors have historically comprised the bulk of convenience and fast-food provision, emerging evidence suggests that, in many countries, large corporations now play an increasingly prominent role in fast-food markets and are becoming more embedded in industrial food supply chains [26, 27]. Evidence from high income countries (HICs), such as the United States, shows 46% of 12–39-year-olds consume fast food daily, comprising 40.6% of their daily energy intake [28]. Similar trends may be emerging in low- and middle-income countries (LMICs). This transition reflects broader drivers of dietary transitions, shaped by factors like urbanisation, rising incomes, and the transformation of labour markets, including the gendered organisation of work and ongoing inequalities in domestic responsibilities [29, 30]. It is also shaped by changes to the political economy of food systems, including globalisation of food supply chains and the rising power of transnational food corporations. These changes have been facilitated by neoliberal policy reforms including trade liberalisation³, deregulation, and privatisation, that have favoured the expansion of large, for-profit firms [31].

¹ Ultra-processed foods, as defined by the NOVA classification scheme (Group 4) are formulations of ingredients, mostly of exclusive industrial use, that result from a series of industrial processes. These products are designed to be hyper-palatable, affordable and convenient. They are almost exclusively produced by transnational food corporations.

² Refer to Table 1 for a comprehensive glossary of terms and definitions.

³ At the time of writing (April 2025), global trade relations are in flux following recent shifts in U.S. trade policy away from liberalisation and toward increased protectionism.

Table 1 Glossary of key terms and definitions related to market power

Term	Definition
Brand equity	The commercial value that derives from consumer perception of the brand name of a particular product or service, rather than from the product or service itself.
Corporate consolidation	The merging of many smaller companies into a few much larger ones. This often occurs through mergers and acquisitions, where one company buys out or merges with another. The result is fewer, larger companies controlling a greater share of the market and wielding significant power over smaller actors.
Corporate fast-food retailer	A large-scale (and often transnational) corporation that provides mass-produced meals through standardised preparation and rapid service for take-away or delivery. They operate via centralised headquarters that control supply chains, brand strategy, and operations, with outlets either corporation-owned or run by franchisees under strict contractual and brand standards [26].
Corporate-industrial food systems	Large-scale food production and distribution systems that are controlled by corporations. These systems are designed to maximise profit, often at the expense of the environment, workers, and animals [40].
Externalities	Costs that are generated by one party, typically a business, but that are borne by or imposed on others. These can be costs that are both quantifiable (healthcare costs) and not quantifiable (e.g., worker dignity or animal welfare) [14].
Financialisation	The “the increasing dominance of financial actors, markets, practices, measurements and narratives” in society [41]
Intellectual monopoly capitalism	A phase of capitalism whereby intangible assets – especially intellectual property – become central to value creation, accumulation, and corporate power [42].
Private equity firms	An investment management company that buys and manages private companies to increase their value and sell them for a profit.
Publicly listed firm	A firm whose ownership is represented by shares of stock that are traded on a public stock exchange, allowing anyone to buy and sell these shares.
Privately listed firm	A firm whose shares are not publicly traded on a stock exchange, rather, they are held privately by a limited number of owners, founders, or investors.
Royalty payments	A payment made to an owner for the right to use their property/asset. Royalties are often based on a percentage of sales or a fixed price per unit sold.
Shareholder primacy	The belief or view that the sole purpose of a business corporation should be to maximise financial returns for its shareholders or private owners [35, 37].
Shareholder value ratio	The shareholder value ratio reflects the value a company provides to its owners [43].

The transformation of fast food from one dominated by diverse, locally rooted practice to one that is, in many countries, highly globalised and corporate-led warrants greater scrutiny, including the expanding market power and financial strategies of the main actors. Yet, most research on both the nutrition transition and the impacts of corporate power in food systems has focused on other sectors, including meat processors [32–34] and UPF manufacturers [6, 35], with some attention to supermarkets as well [36]. Building on this work, we endeavoured to focus on corporate fast-food as a critical, but underexplored, site of analysis. We investigate how the growth of this sector intersects with the financialisation of food systems, particularly through the rising influence of financial actors, the extraction of value via shareholder returns, and the dominance of shareholder primacy⁴ as a model of corporate governance [37]. Analyses of the financialisation of food systems has shown how financial motives increasingly shape strategic decisions across food supply chains, often at the expense of social, ecological and health outcomes. A key mechanism through which this occurs is through ‘shareholder primacy’, i.e., a doctrine embedded in corporate law and governance norms in many jurisdictions, that a firm’s core purpose is the maximisation of financial returns to shareholders and/or private owners [37].

In this study, we aimed to explore the role of FFRs in food system transformation by understanding the financial drivers behind their expansion. We focus on the *corporate* FFR sector, i.e., large-scale, corporatised, and often transnational firms that operate highly standardised business models, including franchised outlets. We analysed global market trends, financial dynamics, and their potential contributions to dietary and systemic transformations. Specifically, this study had two main objectives: (1) to describe recent trends in corporate FFR sales globally, highlighting key patterns and regional variations; and (2) to examine the financial performance of the corporate FFR sector (including the operationalisation of shareholder primacy), by analysing profitability, shareholder payouts of leading global firms, and the role of private equity involvement. We sought to extend upon existing research on the financialisation of food systems by including the corporate FFR sector alongside food manufacturers and meat processors [38, 39].

Methods

To address the aims of this research, we conducted a quantitative analysis of market and business data. Informed by two critical theories (see below), we used quantitative methods to examine a range of indicators related to the structure and market dynamics of the

⁴ Refer to Table 1 for a comprehensive glossary of terms and definitions.

corporate FFR industry. To do this, we analysed several firm- and market-level metrics, with data sourced from two business and market research databases. Periods of quantitative analysis were based on available data. Table 2 provides an overview of the dimensions examined in this study, including the methods, metrics and data sources used.

Data sources

The market sales data, sourced from Euromonitor International Passport Global Market Information (2023 edition) [44], covered corporate FFR sales from 2009 to 2023 across 54 countries, categorised into six regions and three income levels (detailed in Supplementary Table S1). This dataset also provided information on the market share of both privately and publicly listed FFRs. Financial data, obtained from the Compustat North America and Global database (accessed via Wharton Research Data Services) [45], included revenue, net income, and shareholder payouts (i.e., dividends and share buybacks) made by publicly listed FFR corporations, offering insights into their financial performance and growth (1980–2023). A detailed description of the metrics and data sources used for the analysis is provided in Table 2. Trends over time for the investigated indicators are presented as graphs created using R version 4.2.2 [46].

Guiding theories

This study is grounded in a political economy of food systems approach, which examines actors and interests and broader interplay between political, economic and social structures and distributions of power and resources that

shape food production, distribution, and consumption [49, 50]. We draw on food regime theory to situate contemporary trends in the FFR sector within broader historical transformations in global food systems [51, 52]. Food regime theory demarcates distinct historical formations of agri-food relations; including (1) the *colonial-diasporic regime* (first food regime, 1870–1930 s), (2) the *mercantile-industrial regime* (second food regime, 1950s–70s), and most recently, what is most commonly described as (3) the *corporate food regime* (third food regime, late 1980's–present) [50, 53].

While scholars contest and differ in how they characterise the third regime [54], there is broad agreement that it is exemplified by the consolidation of transnational corporate power, financialisation, and the appropriation of localised food cultures [55]. In this study, we adopted the view that the current food regime can be understood as a financialised corporate food regime, and we use regime theory to situate the global expansion of FFR firms and their consolidation of market power as illustrative of this phase. Through our analysis, we explore how leading FFR firms expand and consolidate power within the global food system, and how their financial logics contribute to broader patterns of dietary transformation, environmental degradation, and social inequality.

We adopted this approach alongside the principles of food sovereignty, not as a specific component of analysis, but as a normative contrast to the dynamics of the corporate food regime. Food sovereignty advocates for local control over food systems and prioritises the rights of people to healthy and culturally appropriate food produced through ecologically sound and sustainable

Table 2 Overview of the metrics and data sources for each dimension of analysis

Level	Methods and metrics	Description	Source	Representation
Industry-level	Quantitative analysis of market share by revenue	Market share data was analysed to identify trends and sector dominance of publicly and privately listed FFRs globally and by region.	Euromonitor International Passport (2023 edition)	Figure 1
		As above, by country income level as per World Bank classifications [47].	Euromonitor International Passport (2023 edition); World Bank	Figure 2
		Identified the leading FFR (both public/private), in each available country in the year 2023, focusing on revenue.	Euromonitor International Passport (2023 edition); World Bank	Figure S1
Firm-level	Quantitative analysis of revenue (using pandas [48])	Identified the top 10 publicly listed FFR firms globally, based on trends in annual revenue from 1980 to 2023.	Compustat North America (accessed via Wharton Research Data Services)	Table 3
	Quantitative analysis of revenue and net profit margins	Calculated net profit margins by dividing net income by total revenue for the top publicly listed firms.	Compustat North America (accessed via Wharton Research Data Services)	Fig. 3a and b
	Quantitative analysis of the shareholder payouts in absolute terms and relative to total revenue (i.e., shareholder value ratio)	Calculated the shareholder value ratio by dividing the monetary value of shareholder payouts (i.e., dividends and share buybacks) by total revenue to assess corporate resource allocation trends (1980–2023).	Compustat North America (accessed via Wharton Research Data Services)	Figure 4

methods [56]. A food sovereignty lens helps to draw attention to the ways in which FFRs, though widely critiqued, might still be challenged or reshaped through food sovereignty principles [26], particularly by resisting corporate concentration, financial extraction, and the commodification of food. Thiemann and Roman-Alcalá (2019) contend that fast-food or convenience food is not inherently problematic, but can be reconfigured in a way that shifts from a model based in shareholder primacy towards one that reflects food sovereignty principles such as fair labour and local control, working with nature, and cultural relevance [26]. In doing so, they add important nuance to critiques of fast food and encourages thinking about how it could be part of more just and sustainable food systems.

By integrating food regime theory and food sovereignty principles, this study critically examines the expansion of large corporate entities in the fast-food industry and its role in shaping diets high in intensively produced ASFs and UPFs [57]. Employing this theoretical foundation, we identified metrics to explore the corporate fast-food sector's role in expanding power, perpetuating global dietary shifts that exacerbate health disparities, environmental degradation, and social inequities. Through this lens, we explored how global FFRs operate within and may be contributing to broader structural trends, highlighting pathways for policy interventions that address the interconnected challenges of public health, ecological sustainability, and social justice. We highlighted how these firms operate in ways that constrain alternatives, entrench inequities, and undermine the viability of more just food futures.

Data collection and analysis

Defining the corporate fast-food sector

We define the corporate fast-food sector as the provision of mass-produced meals for commercial resale, characterised by standardised preparation, rapid service, and products that are often frozen, preheated, or precooked and packaged for take-out or delivery. Large-scale corporations in this sector typically operate through a structured model in which centralised headquarters oversee supply chains, brand strategy, and operations, while individual outlets are either directly owned by the corporation or operated by independent franchisees bound by strict contractual and brand standards.

Our definition of the corporate fast-food sector was informed by Euromonitor's classification of 'limited-service restaurants'⁵, which we adopt as the operational

basis for analysing FFR markets. Euromonitor categorises it as mass-produced food for commercial resale, emphasising quick service. The term encompasses food sold in establishments using frozen, preheated, or pre-cooked ingredients, packaged for take-out. We chose not to include independent consumer foodservice; cafes/bars; full-service restaurants; self-service cafeterias; street stalls/kiosks; and consumer foodservice in leisure, retail, or lodging establishments. This definition is appropriate for our analysis as it aligns with other literature on fast-food [28, 58] and focuses on the unique operational and economic characteristics of the corporatised fast-food sector.

Determining market size

To determine market sizes at the global, regional, and country income groups, we used Euromonitor's food-service value retail selling price (RSP) data (i.e., sales at end price to consumer) for 'limited-service restaurants' across 54 countries. All countries with data available in the Euromonitor database were included in our analysis. Sales data for each country were extracted as USD using 2023 fixed exchange rates and were adjusted to 2023 prices by applying Euromonitor's in-built inflation-adjustment function. Per capita sales were then calculated by corresponding data with World Bank population estimates [59]. The compounding annual growth rate (CAGR) was also calculated for the period 2009–2023, representing the average annual growth rate of FFR sales during this time, consistent with similar analyses [6].

Countries were organised into six geographic regions using Euromonitor's classification scheme (Asia Pacific, Eastern Europe, Middle East Africa, Western Europe, Australasia, Latin America, and North America). Countries were also classified into three income groups (Low-and-Middle income countries – LMIC, Upper-Middle income countries – UMIC, and High-income countries – HIC) based on World Bank world development designations in 2024 [47]. These classifications are shown in Supplementary Table S1.

facilities for consumption on the premises. LSR tend to specialize in one or two main entrees such as hamburgers, pizza, or chicken, but they usually also provide drinks, salads, ice cream, dessert, etc. Food preparation is generally simple and involves one or two steps, allowing for kitchen staffs generally consisting of younger, unskilled workers rather than professional chefs. Other key characteristics include: • A standardised and restricted menu • Tight individual portion control on all ingredients and on the finished product • Individual packaging of each item • Counter service • For Chained Limited Service Restaurants, chained and franchised operations which operate under a uniform fascia and corporate identity • Any outlet offering full-service restaurant seating with servers would be excluded, even if delivery service is available." (<https://www.euromonitor.com/definitions>).

⁵ "Limited-Service Restaurants (LSR) combine fast food and 100% home delivery/takeaway outlets. These outlets offer limited menus with items that can be prepared quickly. Customers typically order, pay, and pick up their order from a counter, though some outlets can have limited table service. This also includes 100% Home Delivery/Takeaway which provide no

Identifying and analysing the leading FFR corporations

After determining FFR market sizes, we analysed the top 10 firms in the global FFR industry by annual revenue in five-year intervals from 1980 to 2020, as well as the year 2023, using data from the Compustat North America and Global databases. Throughout the manuscript, we use the term ‘firm’ to refer inclusively to the diverse organisational forms in the sector. In some cases, though, we chose to use more specific terms (e.g., publicly listed corporation, private equity fund, master franchisee). These datasets cover firms listed on stock exchanges in over 80 countries, with records dating back to 1950 [60]. Industry definitions for FFRs were applied based on the North American Industry Classification System (NAICS) (codes 722513/722511) and, where relevant, the Global Industry Classification Standard (GICS) (code 25301040). Explanations of the included codes are in Supplementary Table S2. While code 722,511 pertains to full-service restaurants, we identified several relevant fast-food corporations (such as Wendy’s Co. or Yum! China Holdings) under this category. To address this overlap, we conducted a manual review of firms categorised under both 722,513 and 722,511, excluding those that did not meet our specified definition of fast-food restaurants. We also manually excluded FFRs that did not primarily retail meals containing both meat and UPFs, such as Starbucks Corp and Dairy Queen. As the Euromonitor definition of fast-food retailers is more granular than the Industry Classification System codes, (e.g., it explicitly excludes full-service outlets with servers and emphasises standardised food preparation), we manually reviewed the included industry classifications to ensure non-FFRs (such as casual dining firms, Red Lobster Co. and Cheesecake Factory) were excluded from the analysis.

For an additional analysis of firm profitability and assets, financial data from all publicly listed corporations were extracted from Compustat North America (where

all the leading FFR firms are both headquartered and listed). For consistency in currency conversion over the analysis period, only financial data reported in currencies for which the US Federal Reserve Bank provides publicly accessible nominal exchange rate data were included. These values were subsequently adjusted with GDP deflator data from the World Bank [61] and converted into real values (2023 prices) for analysis. Net profit margins for each FFR were calculated by dividing net income by company revenue. The ‘shareholder value ratio’ was calculated by dividing the total monetary value of shareholder payouts by company revenue. It is a useful proxy for the available funds (excluding debt) that corporations can distribute among their ‘stakeholders,’ including workers, shareholders, governments, and suppliers [48].

To analyse private equity ownership of FFRs, the top 15 privately owned FFRs were first identified using Euromonitor International, ranked by market share in the year 2023. We applied the same parameters as above for inclusion. Private equity ownership was determined through targeted online searches using Google and by examining the websites of prominent private equity firms known for fast-food investments, such as Roark Capital and JAB Holdings.

Results

Part I – market trends at global, regional and country income levels

Global and regional trends in FFR market growth (2009–2023)

The global FFR market expanded markedly between 2009 and 2023. During this period, global sales increased in real terms from US\$724.7 billion in 2009 to US\$984.6 billion in 2023, representing an overall growth of 35.9% and a CAGR of 2.2% (Fig. 1a, Supplementary Table S3). Per capita sales rose from \$105.2 to \$122.7. This trajectory was briefly disrupted in 2020, largely due to the

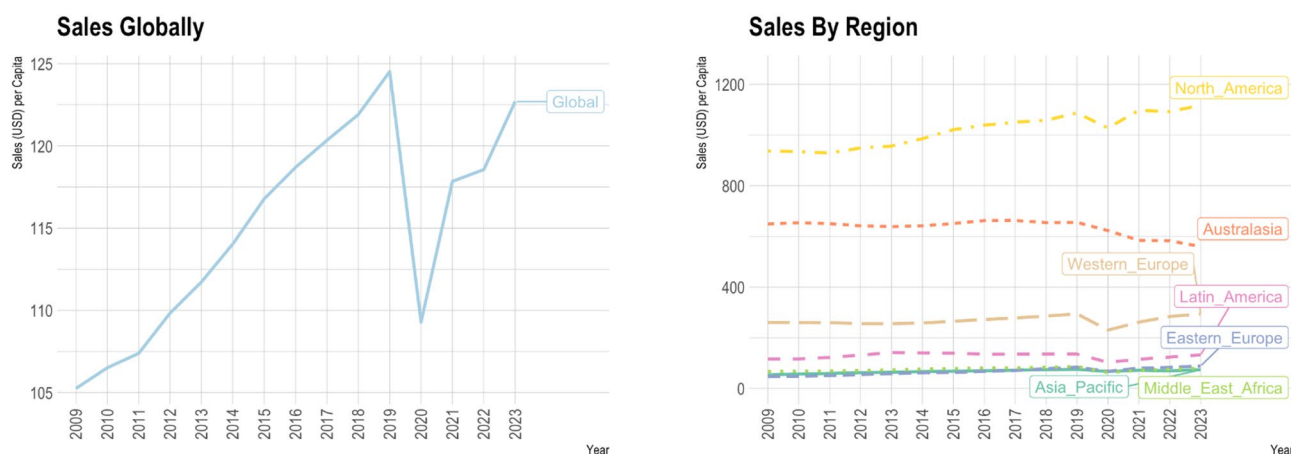


Fig. 1 Per capita FFR sales by (a) globally and (b) region, 2009–2023. Data sourced from Euromonitor International. Per capita sales calculated by corresponding year population estimates from World Bank [59]

COVID-19 pandemic, with per capita sales declining by 12.2% before rebounding in 2021 and returning to a steady upward trend. At a regional level, there were significant variations in sales growth (Fig. 1b, Supplementary Table S4). In more mature markets such as North America, Western Europe, and Australasia, growth appeared relatively stagnated, though these regions collectively continued to demonstrate the highest overall sales. For example, North America – the largest regional market globally – saw sales rise from US\$318.9 billion (\$936.8 p/cap) in 2009 to US\$418.9 billion (\$1,117.0 p/cap) in 2023, an increase of 31.4% (CAGR of 2.0%). This was largely driven by contributions from the United States. Similarly, Western Europe exhibited more restrained growth, with sales rising by 12.7% (CAGR: 0.9%), while sales in Australia declined by 13.7% (CAGR – 1.0%) over the same period.

In contrast, emerging regional markets in the Asia-Pacific and Middle East and Africa demonstrated more substantial growth. Sales in the Asia-Pacific region rose from US\$174.7 billion (\$53.5 p/cap) in 2009 to US\$267.7 billion (\$73.7 p/cap) in 2023, reflecting a 53.2% increase and a CAGR of 3.1%. These were driven largely by large or fast-growing country markets like China, South Korea, and India. China's sales grew from US\$92.4 billion (\$69.4 p/cap) in 2009 to US\$142.5 billion (\$159.8 p/cap) in 2023, a 54.2% increase (CAGR: 3.1%), while India saw an even sharper rise, from US\$2.34 billion (\$1.91 p/cap) to US\$5.19 billion (\$4.65 p/cap), a 121.8% increase (CAGR: 5.9%). South Korea experienced

a similar increase, US\$9.45 billion (\$191.7 p/cap) to US\$20.5 billion (\$238.1 p/cap), a 116.9% increase (CAGR: 5.7%), representing a large proportion of overall sales in the region.

The Middle East and Africa also experienced substantial growth, with sales rising from \$US24.6 billion (\$66.58 p/cap) to \$US39.2 billion (\$79.75 p/cap), increasing by 59.3% (CAGR: 3.4%), with much of this growth attributable to the United Arab Emirates (UAE) and Saudi Arabian markets. For example, the UAE saw sales increasing from US\$1.78 billion (\$221.7 p/cap) in 2009 to US\$3.40 billion (\$312.4 p/cap) in 2023, a 91.0% increase (CAGR: 4.7%). Latin America saw more moderate growth, with sales increasing from US\$50.9 billion (US\$116.7 p/cap) to US\$65.9 billion (\$132.4 p/cap), an increase of 29.5% (CAGR 1.9%), led predominantly by the Mexican market. Mexico's sales rose from US\$14.8 billion (\$133.07 p/cap) in 2009 to US\$17.9 billion (\$148.53 p/cap) in 2023, a 20.9% increase (CAGR: 1.4%). Despite slower growth, Latin America continued to represent a larger share of FFR sales than both the Asia-Pacific and Middle East and Africa regions throughout the period.

Country income level differences in FFR market growth

Our analysis of the growth of the FFR sector by country income level showed distinct patterns of sales growth (Fig. 2, Supplementary Table S5). Consistent with other analyses of dietary change and corporate expansion in the food sector [6, 62], upper-middle-income countries experienced the largest proportion of per capita

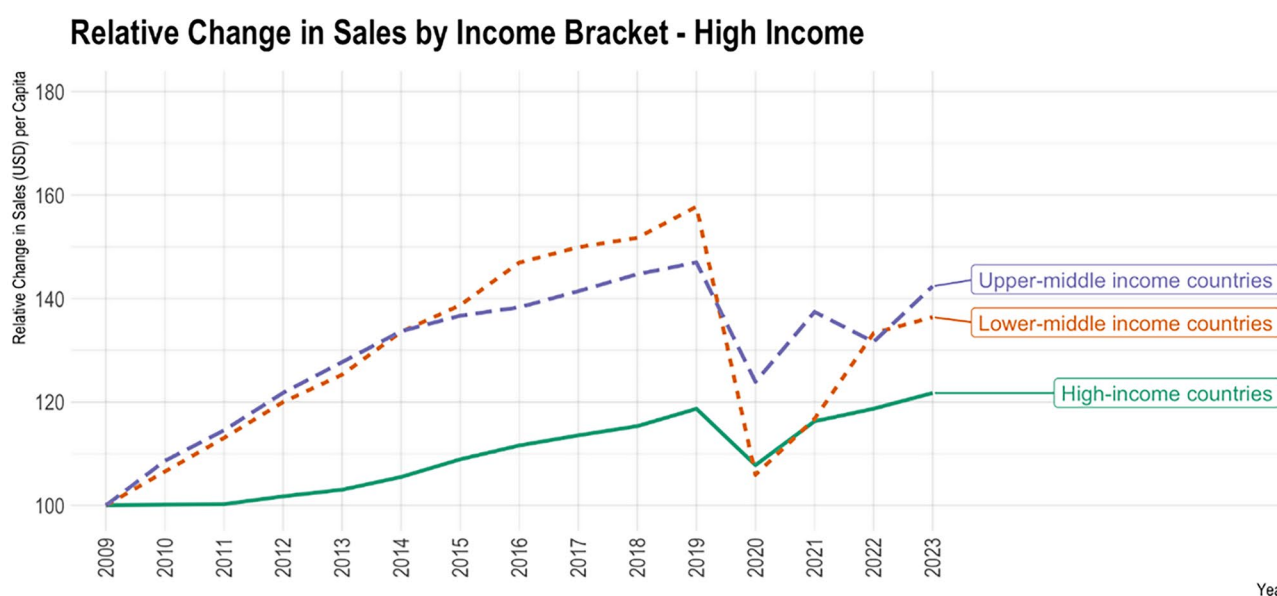


Fig. 2 Relative change in per capita corporate fast-food retailer sales by income level, 2009–2023. Data sourced from Euromonitor International. Income levels were based on World Bank classifications for the year 2023. Per capita sales calculated by corresponding data with World Bank population estimates [59]. Relative change is indexed to 2009 (set to 100); values for subsequent years represent the percentage change in per capita sales relative to 2009 (e.g. 120 = 20% increase)

sales growth, with sales rising by 42.4% (CAGR: 2.6%) over the 15 year period, from US\$61.04 to US\$ 86.89. Lower-middle-income countries followed closely, with an increase of 36.4% (CAGR: 2.2%), from US\$7.69 in 2009 to US\$10.49 in 2023. High-income countries, though showing the least amount of growth, still recorded a considerable increase of 21.7% (CAGR: 1.4%), from US\$431.9 to US\$525.7, reflecting the continued entrenchment of corporate FFR markets in these countries. These countries also had proportionately much higher levels of sales overall.

Part II – financial performance of the leading corporate fast-food retailers

Our analysis showed that, despite sales plateauing in certain markets, some FFR firms were able to maintain relatively stable net profit margins over the analysis period. In this section, we trace the evolution of industry leaders, identifying the current FFR leaders by analysing profitability, revenue, and shareholder payouts over recent decades. This analysis revealed that even with slow sales growth in HICs, leading firms continued

private equity, illustrates how characteristics inherent to the FFR model has increasingly attracted private equity investment.

Top 10 firms by revenue over time & current top firms of each country

In 2023, the ten largest publicly listed FFR corporations by global industry share, measured in terms of revenue, were McDonald's, Yum! China (operating KFC and Pizza Hut in China), Chipotle, Yum! Brands, Restaurant Brands International, Domino's Pizza, and Wendy's Co, Jack in the Box, Shake Shack, and Good Times Restaurants (Table 3). With the exception of Yum! China, all these FFRs are headquartered in the United States. McDonald's maintained its position as the leading company in the global FFR industry for the analysed period (1980–2023)⁶. Franchise operators, such as Arcos Dorados Holdings – a master franchisee of McDonald's restaurants in Latin America – also ranked among the top 10 firms in several years, suggesting the power of brand equity⁷. It is important to note that these data reflect only publicly listed firms. If revenue data for private firms were included in the analysis, it is likely that FFRs such as

Table 3 Leading publicly listed corporations (and country headquarters) in the global FFR industry in terms of revenue, 1970–2023

Rank	1980	1990	2000	2010	2020	2023
1	McDonald's Corp (USA)	McDonald's Corp (USA)	McDonald's Corp (USA)	McDonald's Corp (USA)	McDonald's Corp (USA)	McDonald's Corp (USA)
2	Wendy's Co (USA)	Wendy's Co (USA)	Yum! Brands (USA)	Yum! Brands (USA)	Yum! China Holdings (China)	Yum! China Holdings
3	Carrols Corp (USA)	Jack in the Box (USA)	CKE Restaurants (USA)	Zensho Holdings Co Ltd (Japan)	Chipotle Mexican Grill (USA)	Chipotle Mexican Grill (USA)
4		CKE Restaurants (USA)	Jack in the Box (USA)	McDonald's Holdings Co (Japan)	Yum! Brands (USA)	Yum! Brands (USA)
5		Yoshinoya Holdings Co (Japan)	Yoshinoya Holdings Co (Japan)	Wendy's Co (USA)	Zensho Holdings (Japan)	Restaurant Brands Intl (USA)
6		MOS Food Services (Japan)	Papa Johns International (USA)	Arcos Dorados Holdings Inc (Argentina)	Restaurant Brands Intl (USA)	Domino's Pizza (USA)
7		NPC Restaurant Holdings (USA)	KFC Holdings (Japan)	Restaurant Brands Intl (USA)	Domino's Pizza (USA)	Wendy's Co (USA)
8		Sbarro Inc. (USA)	MOS Food Services (Japan)	Jack in the Box (USA)	McDonald's Holdings Co (Japan)	Jack in the Box (USA)
9		Ringer Hut Co Ltd (Japan)	NPC Restaurant Holdings (USA)	Yoshinoya Holdings Co (Japan)	Arcos Dorados Holdings Inc (Argentina)	Shake Shack (USA)
10		Carrol's Corp (USA)	Carrol's Corp (USA)	Chipotle Mexican Grill (USA)	Papa Johns International (USA)	Good Times Restaurants Inc (USA)

These firms are sourced from Compustat and thus represent publicly listed firms only. Yum! China, established as a spin-off from Yum! Brands in 2016, operates as an independent entity

Franchisers:

- Arcos Dorados Holdings is a master franchisee of McDonald's in Latin America
- Carrols Corp manages franchises of Burger King and Popeyes
- NPC Restaurant Holdings managed franchises of Pizza Hut and Wendy's until 2020
- Autogrill Spa began franchising Burger King and Chick-fil-A in 2007

to make substantial profits. Additionally, an overview of privately owned FFR firms, particularly those backed by

⁶ According to Compustat revenue data, McDonald's Corp has held the top position since 1965 – the same year it became a publicly traded company.

⁷ See glossary in introduction.

CFA Properties (owner of Chick-Fil-A) or Inspire Brands (owner of brands including Subway, Carl's Jr., and Sonic) would have also been included among the leading corporations. Supplementary Figure S1 presents the leading brand (and brand owner) in each analysed country for 2023, including those owned by both publicly listed and privately held firms. In 2023, McDonald's was the leading brand in 39 out of 54 countries, followed by KFC (Yum! Brands), which was the lead in five.

Revenue, profitability and shareholder payouts of leading publicly listed FFR firms over time

Trends in revenue and net profit margins Our data showed that, between 1980 and 2023, the revenue of leading publicly listed FFR corporations exhibited some fluctuations, with some firms experiencing stagnation or decline, particularly around 2020, likely due to the COVID-19 pandemic (Fig. 3a). Consistent with the market data, FFR firms also saw notable revenue dips during this period. Yet these declines did not appear to substantially impact net profit margins (net income relative to revenue), as several brands demonstrated continued, and in some cases, growth in net profit margins during the same period (Fig. 3b). McDonald's demonstrated the most sustained growth, with net profit margins averaging 25.4% from 2015 onwards. Yum! Brands also maintained relatively high profitability, with an average margin of 20.5% over the same period. Yum! China, established in 2016, showed strong but more volatile net profit margins for a young company, averaging 6.6% annually. Restaurant Brands International averaged 13.7% over the 2015–2023 period, while Domino's reported steady profits averaging 10.9%. Chipotle showed more variable margins but maintained an upward trend, averaging 8.2% over 2015–2023. In contrast, Shake Shack and Good Times Restaurants Inc. showed lower and more inconsistent margins overall.

Trends in distribution of shareholder returns Since the 1980's, the leading FFRs – McDonald's, Yum! Brands, and Restaurant Brands International – distributed increasing sums of money to their shareholders relative to their total revenues (Fig. 4). Over this period, McDonald's shareholder value ratio increased from 1.5% in 1980 to 29.8% in 2023, peaking at 57.8% in 2016. Yum! Brands rose from 0% in 1995 to 34.6% in 2023, with an exceptional peak of 68.4% in 2018—surpassing McDonald's at this time. Restaurant Brands International climbed from 10.0% in 2015 to a high of 69.7% in 2017, before falling to 7.1% in 2023, when no dividends were recorded. Wendy's displayed a varied but generally declining trend in shareholder pay-

outs after peaking at 62.6% in 2015⁸; by 2023, its payout ratio had dropped to 8.9%.

Other firms displayed lower or more variable ratios, perhaps influenced in part by their dividend policies. Jack in the Box, for instance, did not issue dividends from 1990 to 2014, which kept its payout ratios low during this period, but subsequently reached a peak of 43.6% in 2018. Both Chipotle and Shake Shack have consistently opted not to issue dividends, resulting in low payout ratios – 6.7% and 0.3% in 2023, respectively – that may reflect a focus on reinvestment over shareholder distributions [63, 64].

Discussion

This study aimed to explore the global expansion and financial dynamics of the corporate FFR sector, with a particular focus on the financial performance of the leading publicly listed corporations. We observed several key trends. Firstly, FFR sales in HICs are stagnating, likely associated with saturated mature markets [65], whereas sales in lower- and upper-middle income countries demonstrated moderate to high levels of growth. The patterns observed align with analyses of sales growth in other segments of the food industry [6, 58]. Second, we found that the global FFR market is largely dominated by U.S.-based firms, with several key players consistently vying for the top positions in most markets, irrespective of region or country income level. Third, we identified that despite declining sales growth in HICs and overall revenue fluctuations throughout the years, these dominant firms have maintained relatively high net profit margins.

While part of this may be due to their expanding presence in LMICs, Fig. 3a shows that overall revenue for the leading firms remained variable. However, our analysis also demonstrated that net profit margins and shareholder payout ratios were comparatively stable, indicating that profitability was less affected by revenue fluctuations. Examining the financial trends of the global FFR industry highlighted how these firms have continued to extract substantial profits under such conditions. These patterns reflect core dynamics of the corporate food regime, in which transnational firms consolidate power, expand into new markets, and maintain profitability.

Global FFR growth: Regional expansion patterns under different political economies

Regional patterns of FFR growth revealed the most rapid expansion in the Asia-Pacific, Latin America, and the Middle East, with US-based FFRs dominating market shares in key countries of growth such as China, India,

⁸ In 2014, Wendy's Co sold 418 company-owned stores as part of a refranchising strategy, which may explain the spike in shareholder returns for 2015.

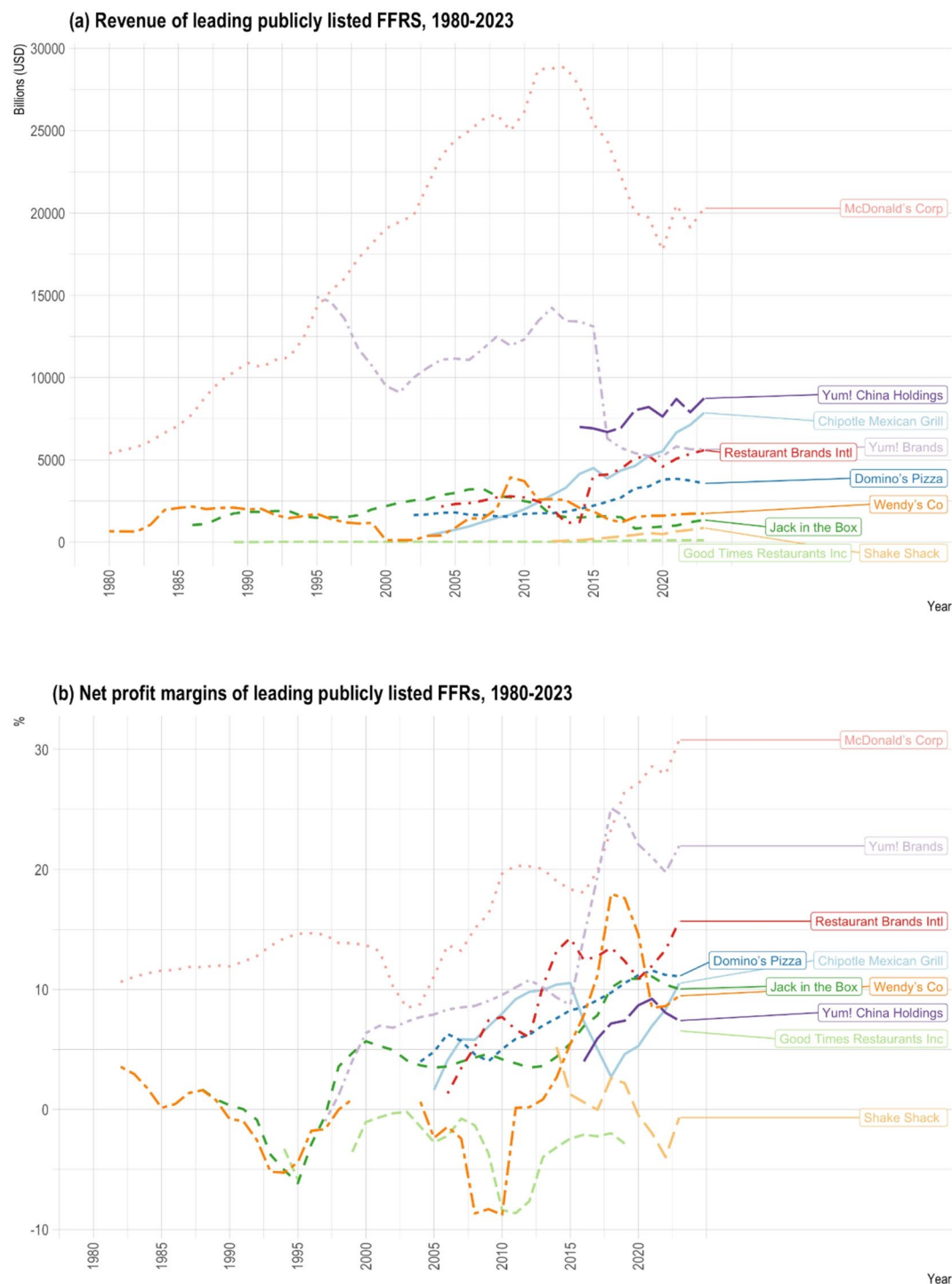


Fig. 3 Trends in profitability of leading publicly listed FFRs based on **(a)** revenue, and **(b)** net profit margins, 1980–2023. Data sourced from Compustat North America, accessed via Wharton Research Data services. Net profit margins = net income/revenue. The data represented in the graphs is shown as a 3-month moving average. Figure 3b includes breaks in reported net profit margins due to substantial outliers associated with data anomalies. Wendy's Co shows a pronounced spike in 2000 (505.8%), likely because Compustat attributed the net income of Triarc Companies Inc.—which became Wendy's parent company in 2008—as a corporate predecessor. Similarly, Good Times Restaurants Inc. displays extreme values for 1989–1991, 1996, and 2020, likely reflecting inconsistencies in reporting or data coverage common among smaller firms. This approach is used to smooth out short-term fluctuations and highlight longer-term trends or patterns in the data

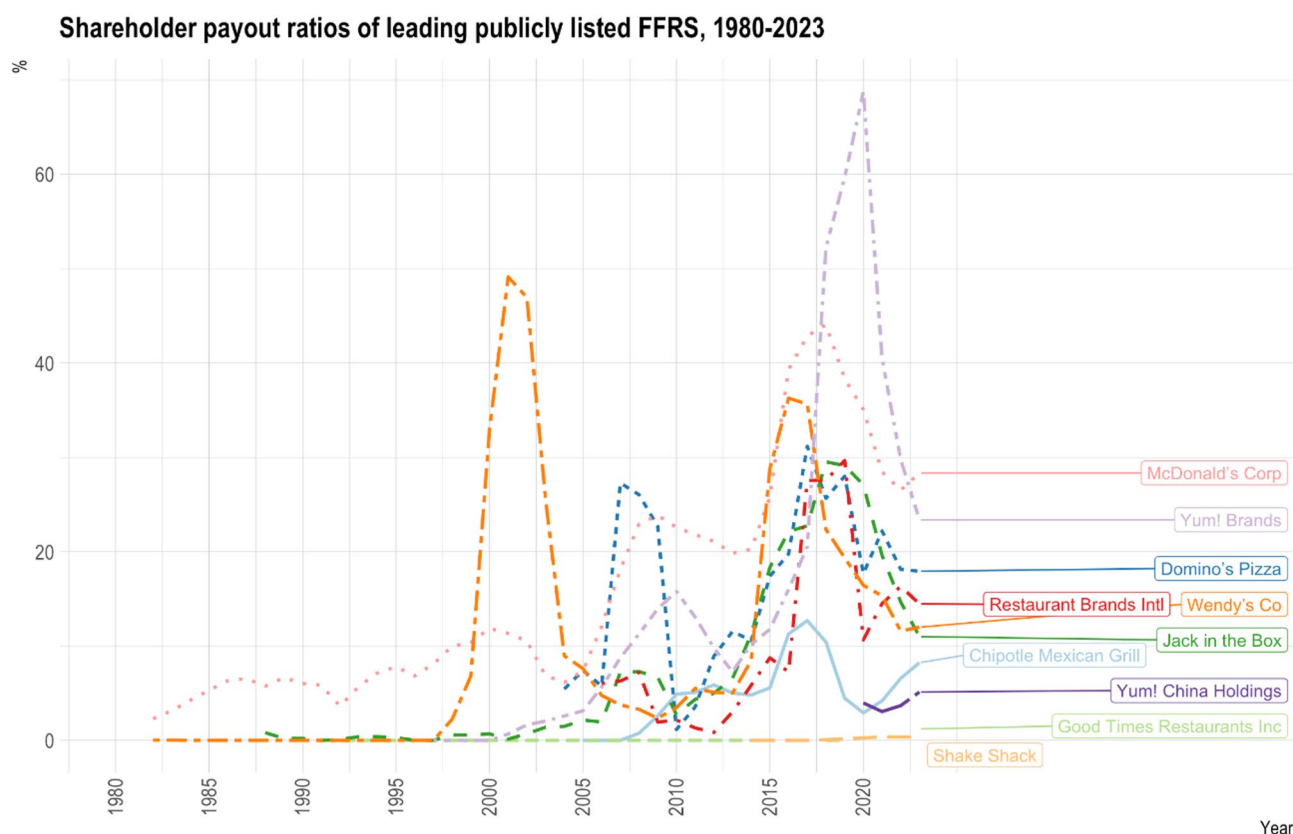


Fig. 4 Shareholder returns paid out relative to revenue (shareholder value ratio) of leading FFRs, 1980–2023. Data sourced from Compustat North America, accessed via Wharton Research Data services. Total shareholder payout ratios were calculated as the sum of dividends paid and share repurchases, divided by company revenue. Share repurchase data from Compustat may include data on purchase of preferred stock. The shareholder payout ratio data for Good Times Restaurants was excluded from this figure due to an outlier ratio between 1990–1995, which disproportionately skewed the scale of the figure and obscured meaningful trends across other firms

South Korea, Mexico, the United Arab Emirates, and Saudi Arabia. The expansion of fast-food markets in the Asia-Pacific region in particular reflects key dynamics of the nutrition transition and imbuing of corporate food systems in these parts of the world. For example, China's FFR sales grew by 54.2% (CAGR 3.1%) between 2009 and 2023, far outpacing other markets in the region (with the exception of India). KFC, in particular, has capitalised on China's increasing preference for chicken over other meats and is now deeply embedded in China's urban food culture as the most dominant fast-food chain [58] (Supplementary Figure S1). KFC was one of the first FFRs to enter the Chinese market, establishing a joint-venture in 1987 with 60% ownership held by KFC, 27% by the Beijing Municipal Bureau of Culture and 13% by Beijing Food Production [66]. It was also one of the first FFRs to develop a localisation strategy in the region, tailoring their menu offerings to Chinese cultural preferences [67]. Fried chicken, in particular, is more closely aligned with traditional Chinese culinary practices than hamburgers, which may partially explain how KFC has been able to outperform McDonald's [68]. A recent survey found that

Chinese consumers are more likely to dine at KFC restaurants and spend more time there than American consumers [69].

In the Middle East and Latin America, increasing FFR market growth was observed to be strongest in Saudi Arabia, the UAE, and Mexico. These countries share several features that likely facilitated fast-food expansion. First, all have undergone nutrition transitions, marked by rising consumption of ultra-processed and convenience foods, alongside increasing incomes and rates of urbanisation [6]. In Saudi Arabia and the UAE, rising interest for convenient, Western-branded food has aligned with economic diversification strategies – such as Saudi Vision 2030 – which has played an active role in attracting foreign investment in retail and food service [70]. Both countries have also supported franchising and private-sector growth through land use policies and relaxed foreign ownership rules in a range of sectors, including food service [71, 72]. In contrast, many scholars have speculated the North American Free Trade Agreement playing a key role in integrating Mexico into US food and investment markets, including through US agribusiness

investment [73–75]. This penetration has increasingly shaped urban foodscapes by embedding the presence of transnational FFR and UPF firms in cities that have traditionally been serviced by small-scale vendors [76], a substantial cut to national and urban food sovereignty.

FFR growth was observed to be strongest in China, India, and South Korea, contrasting with countries like Japan, which saw more limited growth between 2009 and 2023. A range of factors may explain this. From a policy perspective, China has allowed foreign corporate activity but under relatively strict conditions that maintain state oversight [77, 78]. For example, McDonald's has functioned in China through joint ventures with firms CITIC Group (a Chinese state-owned enterprise) and Carlyle (a US private equity firm), with CITIC maintaining a controlling 52% stake until 2023 [79]. In contrast, while Japan has similarly introduced foreign FFRs into their food systems, it has maintained stronger protections for domestic food service industries, which has limited foreign FFR expansion relative to China [80]. For example, the Japanese government provides a range of subsidies and support to local small- and medium-sized enterprises [81]. This strategy may align with Japan's desire to maintain its culinary culture and a cultivate a preference for local cuisine. In this light, while McDonald's is the leading FFR brand in Japan, the country still maintains a competitive domestic FFR market, which may moderate the overall market share of transnational FFR firms [80]. These contrasts underscore how corporate food regimes can manifest in different ways under different political economies, and how different forms of state-market relations interact with the power of corporate FFRs.

How FFRs service shareholder primacy: from food service retailers to real estate moguls

Franchising has emerged as a key profit-maximising strategy through which corporate FFRs reduce operational costs and risks, enabling new forms of value extraction that align with the broader logics of financialisation and the consolidation of the corporate food regime. Many large FFRs have adopted a highly franchised business model, shifting the burden of direct operational expenses onto franchisees while maintaining centralised control over the brand. This arrangement allows franchisees to leverage the power of the brand to generate sales of the franchisor in exchange for royalty payments and a portion of their profits; although the benefits to franchisees are not always realised, as they often face significant financial pressures and operational constraints [82, 83]. For example, following Yum! China's spin-off from Yum! Brands in 2016 (a move driven by broader strategic goals in the region as well as challenges in the Chinese market, including previous food safety scandals), 93% of Yum! Brands' locations were franchised globally. This figure

increased to 98% by 2018 as part of Yum! Brands' broader strategy to reduce operational costs and risks, as well as revenue recovery from the spin-off [84]. The shift allowed the company to focus on generating revenue through franchise fees and rent, creating a more 'asset-light' business model [85]. This mirrors broader transformations within the corporate food regime, whereby firms increasingly profit not just from producing food, but also from controlling the infrastructures and land through which food is distributed, branded, and consumed. Such strategies see corporate FFRs transitioning away from being solely food service providers, with profits being increasingly derived from rent collection rather than direct food sales [86].

McDonald's, with a global brand valuation of approximately \$200 billion USD, sees average franchise investments in the US ranging between one to two million dollars [87]. When McDonald's faced declining revenues in 2015, it reportedly financed its stock buy-back program by increasing its reliance on franchising (alongside other cuts in 'general and administrative expenses') [88]. These strategies exemplify how shareholder primacy is operationalised within the corporate food regime, as financial returns to investors are prioritised over reinvestment in productive or social functions. The focus on brand value aligns with related trends in "intellectual monopoly capitalism" [42], where companies rely on intellectual property like trademarks and brands to maintain high market values even if their sales revenue does not grow at the same rate [89, 90]. Such a trend has not been without negative consequences. Franchising places significant pressure on workers, especially in terms of wages and working conditions [91]. Research shows that in advanced capitalist economies (like the United States), the franchising model incentivises franchisees to minimise wages and extract greater productivity from employees [87], raising legal and ethical concerns about potential labour exploitation [86].

Our analysis reveals increased expansion of FFR markets in LMICs, which introduces additional complexities. Emerging FFR markets in LMICs are attractive to global franchisors, yet the regulatory frameworks that govern franchising in these contexts are often underdeveloped [92]. This lack of regulation can lead to unequal contracts between franchisors and franchisees, creating information gaps that result in unfair practices [92]. Smaller, local franchisees are especially at risk, as they may struggle to navigate or challenge these imbalances, particularly in places where legal protections are weak or biased in favour of large multinational companies [92, 93]. Such dynamics pose substantial challenges to food sovereignty, as small- and medium-sized local actors lose control to transnational firms, who consolidate their dominance through both financial and contractual mechanisms. In

this context, franchising becomes not only a business strategy, but a mode of governance that reconfigures local food economies in ways that are often incompatible with healthy, sustainable and equitable food systems.

The intensification of shareholder primacy in the corporate FFR sector

Our analysis revealed that since the 1980's, the leading FFRs – McDonald's, Yum! Brands, and Restaurant Brands International – distributed increasing sums of money to their shareholders relative to their total revenues. We found a peak in McDonald's shareholder value ratio in 2016 (57.8%), which is likely attributable to the company's record US\$11.2 billion share buyback program it implemented that year [94]. Similarly, Yum! Brands' ratio increased markedly from 2018, peaking at 68.4%, shortly after a major US\$5.4 billion buyback in 2016 and during a period of continued capital returns [95]. Share buyback programs are a financial strategy whereby companies repurchase their own shares from the market, thereby reducing the number of outstanding shares. They are often used to increase earnings per share, signal confidence in future performance, and return surplus capital to shareholders in lieu of dividends [96]. They also function to increase share prices in the short term, thereby enhancing capital gains for shareholders and inflating executive compensation tied to stock performance [97]. In the context of McDonald's and Yum! Brands, these buybacks coincided with major organisational transformations – including refranchising, cost-cutting, and structural separation (in Yum's case, the spin-off of Yum China) – suggesting that buybacks may have also been used to manage investor perceptions and reinforce a commitment to shareholder returns during periods of strategic transition.

These escalating shareholder returns – facilitated by mechanisms like buybacks – reflect and reinforce the growing influence of financial actors in the sector. Similar to the agricultural input and meat processing sectors, the global FFR industry is characterised by the considerable involvement of institutional investors and private companies (i.e., companies that are not publicly listed) [98]. The role of such financial actors in shaping the strategic orientation of the global FFR sector warrants greater attention. Similar to other food system sectors, institutional investors – including asset managers and pension funds – are major shareholders in publicly listed FFR firms [35], bolstering the shareholder primacy model in the sector. At the same time, FFRs have become increasingly attractive to private equity, due in part to its franchising model and opportunities for rapid cost-cutting [99]. Private equity's short-termist logic – often reliant on debt-fuelled acquisitions and aggressive restructuring – may exacerbate financial instability and undermine labour

and sustainability outcomes. A leading example is Roark Capital, which has become the dominant private equity firm in the global FFR sector [100], owning 60 different franchise and multi-unit brands that, collectively, generate approximately US\$24 billion in revenue across 27,000 locations in 78 countries [101]. The implications of common and private ownership merit further investigation but are beyond the scope of this paper.

Implications for food systems: governance, public health and equity, cultural shifts, and environmental impacts

The emphasis on shareholder primacy in corporate FFRs has significantly accelerated their global expansion, cementing many U.S.-based corporate FFRs as the dominant provisioners of fast-food (Supplementary Figure S1). While corporatisation of food systems has been the norm in advanced capitalist economies like the United States and the United Kingdom for some time [26], our analysis indicates that this trend is rapidly unfolding in LMICs, consistent with the broader nutrition transition phenomenon. Large FFR corporations play a role in displacing traditional diets, meals, food services, and cultures – as well as the capacity for alternative fast-food industries to emerge and flourish [102]. With a focus on convenience and meals centred around intensively produced meat⁹ and ultra-processed foods, FFRs not only reflect, but also enable, capitalist transformations in society, reducing time and energy dedicated to food preparation and consumption [103]. In this sense, corporate fast food can be seen as both a manifestation and a driver of the corporate food regime [30, 104].

The displacement of traditional meals is particularly concerning in countries outside the global North. As urbanisation intensifies, the market for convenience foods grows, and FFRs – predominantly originating from the United States – actively adapt their offerings to align with local cultural preferences, a process known as 'glocalisation' [105]. For instance, in South Korea, China, and Saudi Arabia, our findings highlight significant market expansion driven by KFC and McDonald's, which have incorporated regional dietary preferences such as increased chicken and pork consumption (except in Saudi Arabia, where pork is prohibited) [106]. Franchising facilitates this 'glocalisation' by enabling FFRs to acquire localised knowledge about consumer preferences, supply chains, and business practices, enhancing their competitive edge [58].

However, corporate FFRs also displace local convenience food vendors in LMICs, many of which are

⁹ It should be noted that in recent years, some FFRs have responded to global recommendations to reduce meat intake by offering plant-based meat alternatives. However, much of these options are ultra-processed and represent a strategy that maintains the underlying logics of industrial production and corporate control.

small- and medium-sized enterprises offering traditional foods that offer more diverse and culturally embedded food options, often distinct from the standardised, ultra-processed and meat-intensive offerings of corporate FFRs [26]. While corporate FFRs are not the only vendors of unhealthy and unsustainable foods, their scale, ownership structures, and growth strategies may afford them a disproportionate capacity to shape global dietary patterns compared to small-scale vendors [107, 108]. When corporate FFRs headquartered in HICs expand into LMICs, this dynamic could be considered a form of neo-colonialism, as it often encroaches on local food systems and pushes local alternatives out of the market [109, 110]. Such displacement not only has implications for food cultures, but raises concerns regarding the sustainability and health impacts of shifting from traditional diets to those reliant on ultra-processed foods and industrially produced meat. This is because the expansion of corporatised FFRs, with their reliance on global supply chains, intensifies the demand for intensively produced meat, UPFs, and their associated inputs – such as commodity crops and plastics [111] – contributing to deforestation, biodiversity loss, and increased greenhouse gas emissions. The long-term environmental and public health consequences of allowing corporate FFRs to dominate local markets and displace more sustainable, traditional food systems warrants critical consideration.

Potential policy responses

This study highlights the increasing expansion and financialisation-driven restructuring of the global fast-food industry, suggesting that policy responses are best targeted toward the structural factors that allow these corporations to proliferate in food systems. Much of the policy proposals from the public health sector have focused narrowly on piecemeal and individual-level options, such as providing nutrient information or kilojoule labelling on menus [112]. Some food environment-based policies include zoning policies¹⁰ that restrict FFR placement in relation to schools [113]. However, very few policy responses have attempted to tackle the deeper economic and political structures that enable FFRs to expand and exert power. One key policy intervention is to dismantle government subsidies that disproportionately benefit large agri-food corporations (such as agricultural subsidies toward commodity crops like corn and soy), thereby leveling the playing field for small- and medium-sized enterprises, which often provide more diverse and culturally relevant food options [26, 102]. Such an approach may help to regulate, and even actively

reshape, the supply chains that feed into these industries, particularly in relation to the sourcing of intensive meat and dairy and ultra-processed foods that the corporate FFR industry relies on. Moreover, leveraging competition policies to address the concentration of corporate power, particularly in LMICs, could play an important role in supporting small- and medium-sized enterprises and addressing market dominance [114, 115]. Strengthening competition law could also help to curb private equity-led consolidation, where firms use ‘roll-up’ strategies to acquire and merge smaller competitors, reducing market competition [116]. Regulatory bodies, such as the U.S. Federal Trade Commission under the Biden administration¹¹, have begun scrutinising these tactics, recognising their role in entrenching corporate dominance across various industries, including fast-food [117]. These tools could represent critical leverage points for healthy and sustainable food system advocates, who may find opportunities to challenge industry power by aligning with broader efforts to curb excessive financial speculation and corporate consolidation.

Environmental and labour regulations, such as due diligence laws that require greater accountability in sourcing and production, could also limit the expansion of corporate food supply chains that sustain this particular fast-food model [118, 119]. The way this might work in practice and its impact on food supply chains warrants further investigation. Moreover, strengthening social and income protections can also reduce the socio-economic barriers that drive dependence on cheap fast food, improving access to sustainable and healthy diets for all [120].

Targeting the franchising model and private equity involvement in FFRs through stronger labour protections and franchisee rights may prevent exploitative practices, especially in LMICs where regulatory frameworks are less developed. Additionally, strengthening regulations on private equity – such as disincentivising leveraged buyouts and closing tax loopholes that benefit these firms – could curb the most extractive financial practices in the sector. Regulating private equity’s role in the sector could further safeguard local economies and workers from the negative impacts of financialisation, contributing to a more resilient and equitable food system [121]. From a food sovereignty perspective, these protections could help safeguard the flourishing of local food economies, reduce dependency on transnational corporate actors, and support more democratically governed food systems.

¹⁰ In 2025, McDonald’s overturned multiple UK council decisions to restrict outlet placement near schools, using legal threats and corporate health claims to undermine local authority efforts (Borland, 2025).

¹¹ It is not known whether the Federal Trade Commission, under the leadership of the Trump Administration, will continue this focus in 2025 and beyond.

Strengths and limitations

This study provides a comprehensive analysis of the global FFR industry, leveraging large datasets to identify key market trends, corporate strategies, and financial dynamics. A major strength of this work is its focus on transnational corporations, allowing for a detailed examination of how leading FFR firms operate across diverse markets. The study also contributes to the broader political economy of food literature on corporate food systems by documenting the increasing penetration of FFRs in LMICs and the financial practices reshaping the sector. Additionally, by using standardised financial and sales data, this analysis offers a robust, comparative perspective on FFR market expansion and corporate profitability across different regions.

However, several limitations should be acknowledged. First, as with any secondary data analysis, findings are constrained by the availability and quality of existing data sources. While our dataset includes 54 countries, which is a substantial proportion, it nonetheless represents only 27% of the world's nations, limiting its global coverage. Furthermore, although we analyse long-term trends, we do not explicitly account for the broader economic, political, or public health contexts—such as financial crises, trade policies, or the COVID-19 pandemic—that may have influenced these trends.

A key limitation is the lack of granular data on FFR sales by meal or product categories. This restricts our ability to directly quantify changes in the sales of UPFs and meat, despite evidence that these products form a substantial part of corporate fast-food offerings [22].

Another methodological constraint is the discrepancy between our two main data sources. Euromonitor data covers 2009–2023, while Compustat data extends from 1980 to 2023, creating differences in the temporal scope of our analysis. While both datasets provide valuable insights, this discordance means that financial trends observed in Compustat over the long term could not be directly compared with market sales trends captured by Euromonitor in more recent years. It was also beyond the scope of the analysis to verify data extracted from Compustat, which may contain instances of missing, duplicated, or inconsistent figures, particularly for firms listed outside North America. Moreover, currency exchange rate fluctuations may have influenced some observed trends.

Future research could provide a more nuanced analysis of the financial strategies employed by the corporatised FFR sector, with a particular focus on the leading firms. Additionally, examining how FFRs intersect with other key actors in the food supply chain, such as meat processors and ultra-processed food manufacturers, as well as other financial actors, such as asset managers, could illuminate the broader corporate dynamics shaping

corporate-industrial food systems. Finally, understanding how these financial and operational strategies impact the intersecting outcomes of public health, equity, and environmental sustainability, could inform future holistic policy interventions.

Conclusion

This study highlights the increasing expansion of corporate FFRs globally, particularly its rapid growth in LMICs. Our analysis shows that while sales in HICs are stagnating, leading firms continue to extract substantial profits through financial strategies that insulate them from revenue fluctuations. The dominance of US-based corporations in nearly all markets underscores the homogenisation of the corporate fast-food sector, shaping food environments and limiting the space for alternative fast-food industries. The emerging popularity of online food delivery services has likely exacerbated these networks, and therefore, consumption trends [122, 123], particularly as corporate FFRs increasingly form partnerships with these platforms [124]. Today, most fast-food production and consumption operate as elements of the corporate food regime in many countries in both the global North and South [26, 52]. Our findings align with broader research on the nutrition transition, demonstrating how corporatised FFRs displace traditional food cultures and services, reinforcing a globalised food system centred on convenience, intensively produced meat, and UPFs – with detrimental implications for public health, our ecological systems, and justice for workers, farmers, and non-human animals. Challenging such expansion is therefore not only a matter of health or sustainability, but also of democratic control over food systems.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12992-025-01158-9>.

Supplementary Material 1

Acknowledgements

N/A.

Author contributions

KS, BW, PB, and GS contributed to the study design. KS collected the data. KS and TB analysed the data, and TB produced the graphs and figures. KS drafted the manuscript, with critical reviews and edits provided by BW, TB, TS, PB, and GS. All authors read and approved the final manuscript.

Funding

KS and TS are supported by Deakin University Postdoctoral Research Fellowships. PB is supported by a Future Fellowship awarded by the Australian Research Council (# FT220100690, and a Horizon Fellowship awarded by the University of Sydney. GS is supported by a Fellowship from the Australian National Health and Medical Research Council (NHMRC, 2021/GNT2008535) and receives grant funding from the Victorian Health Promotion Foundation (VicHealth), the Australian Research Council (ARC), the National Heart Foundation of Australia, and Canadian Institutes of Health Research (CIHR).

Data availability

No datasets were generated or analysed during the current study.

Declarations**Ethics approval**

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹Global Centre for Preventive Health and Nutrition, Institute for Health Transformation, Faculty of Health, Deakin University, Geelong, Australia

²Biostatistics Unit, Faculty of Health, Deakin University, Geelong, VIC, Australia

³Sydney School of Public Health, Faculty of Medicine and Health, University of Sydney, Sydney, NSW, Australia

Received: 2 May 2025 / Accepted: 7 October 2025

Published online: 31 October 2025

References

1. FAO. Sustainable food systems - concept and framework. Rome: FAO: Food and Agriculture Organization of the United Nations; 2018.
2. IPES-Food. Food from somewhere: Building food security and resilience through territorial markets. Brussels: IPES-Food: IPES-Food; 2024.
3. Stephens P, Clapp J, Isakson R. Financialisation and sustainable diets. Routledge handbook of sustainable diets. Routledge; 2022. pp. 442–53.
4. Lacy-Nichols J, Scrinis G. Health and the corporate agri-food system. The Routledge Handbook of the Political Economy of Health and Healthcare: Routledge; 2024. p. 143–53.
5. Anastasiou K, Baker P, Hendrie GA, Hadjikakou M, Boylan S, Chaudhary A, et al. Conceptualising the drivers of ultra-processed food production and consumption and their environmental impacts: a group model-building exercise. *Global Food Secur.* 2023;37:100688.
6. Baker P, Machado P, Santos T, Sievert K, Backholer K, Hadjikakou M, et al. Ultra-processed foods and the nutrition transition: Global, regional and National trends, food systems transformations and political economy drivers. *Obes Rev.* 2020;21(12):e13126.
7. World Health Organization. Red and processed meat in the context of health and the environment: many shades of red and green. Information brief Geneva: World Health Organization; 2023.
8. Anastasiou K, Baker P, Hadjikakou M, Hendrie GA, Lawrence M. A conceptual framework for Understanding the environmental impacts of ultra-processed foods and implications for sustainable food systems. *J Clean Prod.* 2022;368:1133155.
9. Djekic I, Tomasevic I. Environmental impacts of the meat chain—Current status and future perspectives. *Trends Food Sci Technol.* 2016;54:94–102.
10. Kastner T, Chaudhary A, Gingrich S, Marques A, Persson UM, Bidoglio G, et al. Global agricultural trade and land system sustainability: implications for ecosystem carbon storage, biodiversity, and human nutrition. *One Earth.* 2021;4(10):1425–43.
11. Kirkwood JK, Sainsbury AW, Bennett PM. The welfare of free-living wild animals: methods of assessment. *Anim Welf.* 1994;3(4):257–73.
12. Lo J, Jacobson A. Human rights from field to fork: improving labor conditions for food-sector workers by organizing across boundaries. *Race/Ethnicity: Multidisciplinary Global Contexts.* 2011;5(1):61–82.
13. Fraser D, MacRae AM. Four types of activities that affect animals: implications for animal welfare science and animal ethics philosophy. *Anim Welf.* 2011;20(4):581–90.
14. Sievert K, Lawrence M, Parker C, Baker P. How power in corporate-industrial meat supply chains enables negative externalities: three case studies from Brazil, the US, and Australia. *One Earth.* 2024.
15. Fardet A, Rock E. Ultra-processed foods and food system sustainability: what are the links? *Sustainability.* 2020;12(15):6280.
16. Willett W, Rockström J, Loken B, Springmann M, Lang T, Vermeulen S, et al. Food in the anthropocene: the EAT–Lancet commission on healthy diets from sustainable food systems. *Lancet.* 2019;393(10170):447–92.
17. FAO. Contribution of terrestrial animal source food to healthy diets for improved nutrition and health Outcomes—an evidence and policy overview on the state of knowledge and gaps. Italy: FAO Rome; 2023.
18. FAO. Ultra-processed foods, diet quality, and health using the NOVA classification system. FAO: Rome: Food and Agriculture Organization; 2019.
19. World Health Organization and Food and Agriculture Organization of the United Nations. What are healthy diets? Joint statement by the food and agriculture organization of the united nations and the world health organization. Geneva: World Health Organization and Food and Agriculture Organization of the United Nations; 2024.
20. Beal T, Gardner CD, Herrero M, Iannotti LL, Merbold L, Nordhagen S, Mottet A. Friend or foe? The role of Animal-Source foods in healthy and environmentally sustainable diets. *J Nutr.* 2023;153(2):409–25.
21. de Araújo TP, de Moraes MM, Magalhães V, Afonso C, Santos C, Rodrigues SS. Ultra-processed food availability and noncommunicable diseases: a systematic review. *Int J Environ Res Public Health.* 2021;18(14):7382.
22. Frank SM, Taillie LS, Jaacks LM. How Americans eat red and processed meat: an analysis of the contribution of thirteen different food groups. *Public Health Nutr.* 2022;25(5):1–10.
23. Ellithorpe ME, Zeldes G, Hall ED, Chavez M, Takahashi B, Bleakley A, Plasencia J. I'm lovin' it: how fast food advertising influences Meat-Eating preferences. *J Health Communication.* 2022;27(3):141–51.
24. Basile AJ, Noshirwani NK, Sweazea KL. 85% of menu items from the Six highest selling fast-food restaurants in the United States are ultra-processed. *Public Health Nutr.* 2025:1–15.
25. Dixon P. Nuggets of gold: further processed chicken and the making of the American diet. University of Georgia; 2024.
26. Thiemann L, Roman-Alcalá A. Fast food sovereignty: contradiction in terms or logical next step? *J Agric Environ Ethics.* 2019;32(5–6):813–34.
27. Dunn KI, Mohr PB, Wilson CJ, Wittert GA. Beliefs about fast food in Australia: a qualitative analysis. *Appetite.* 2008;51(2):331–4.
28. Popkin BM. Does excessive fast-food consumption impair our health? *Am J Clin Nutr.* 2022;116(1):11–2.
29. Popkin BM, Ng SW. The nutrition transition to a stage of high obesity and noncommunicable disease prevalence dominated by ultra-processed foods is not inevitable. *Obes Rev.* 2022;23(1):e13366.
30. Stevano S. Ultra-processed food, depletion, and social reproduction: a conceptual intervention. *Antipode.* 2024.
31. Lawrence G. Re-evaluating food systems and food security: a global perspective. *J Sociol.* 2017;53(4):774–96.
32. Howard PH. Corporate concentration in global meat processing: the role of government subsidies. Workshop on global meat industry and its implications. 2017.
33. Howard PH. Corporate concentration in global meat processing: the role of feed and finance subsidies. *Global meat: social environ consequences expanding meat Ind.* 2019:31–53.
34. Sievert K, Lawrence M, Parker C, Baker P. Understanding the political challenge of red and processed meat reduction for healthy and sustainable food systems: a narrative review of the literature. *Int J Health Policy Manage.* 2021;10(Special Issue on Political Economy of Fo).
35. Wood B, Robinson E, Baker P, Paraje G, Mialon M, van Tulken C, Sacks G. What is the purpose of ultra-processed food? An exploratory analysis of the financialisation of ultra-processed food corporations and implications for public health. *Globalization Health.* 2023;19(1):85.
36. Howard PH. Increasing corporate control: from supermarkets to seeds. In: Konefal J, Hatanaka M, editors. *Twenty lessons in the sociology of food and agriculture.* 2019:117–31.
37. Sjöfjell B, Taylor MB. Clash of norms: shareholder primacy vs. sustainable corporate purpose. *Int Comp Corp Law J.* 2019;13(3):40–66.
38. Howard PH, Ajena F, Yamaoka M, Clarke A. Protein industry convergence and its implications for resilient and equitable food systems. *Front Sustainable Food Syst.* 2021;5:684181.
39. Wood B, Williams O, Baker P, Sacks G. Behind the 'creative destruction' of human diets: an analysis of the structure and market dynamics of the ultra-processed food manufacturing industry and implications for public health. *J Agrarian Change.* 2023;23(4):811–43.
40. Clapp J. The rise of big food and agriculture: corporate influence in the food system. Edward Elgar Publishing. 2022:45–66.

41. Aalbers MB. Corporate financialization. In: Richardson D, Castree N, et al. editors. *The international encyclopedia of geography: People, the earth, environment, and technology*: Wiley-Blackwell. 2017.
42. Sell SK, What. COVID-19 reveals about twenty-first century capitalism: adversity and opportunity. *Development*. 2020;63(2–4):150–6.
43. Hager SB, Baines J. The tax advantage of big business: how the structure of corporate taxation fuels concentration and inequality. *Politics Soc*. 2020;48(2):275–305.
44. Euromonitor International. *Passport global market information database*. London: Euromonitor International; 2023.
45. Wharton Research Data Services. *Compustat North America and global database*. University of Pennsylvania. 2024.
46. R Core Team. R: A language and environment for statistical computing. In: R foundation for statistical computing, editor. Vienna, Austria. 2024.
47. World Bank. *DataBank: world development indicators online*: World Bank. 2024 [Available from: <https://databank.worldbank.org/source/world-development-indicators>]
48. pandas. *Pandas 2.2.3*. NumFOCUS; 2024.
49. Baker P, Lacy-Nichols J, Williams O, Labonté R. The political economy of healthy and sustainable food systems: an introduction to a special issue. *Int J Health Policy Manage*. 2021;10(12):734.
50. McMichael P. Political economy of the global food and agriculture system. *Rethinking food and agriculture*: Elsevier; 2021. pp. 53–75.
51. Friedmann H, McMichael P. Agriculture and the state system. *Sociologia Ruralis*. 1989;29(2).
52. McMichael P. A food regime genealogy. *Critical perspectives in rural development studies*. Routledge; 2013. pp. 129–58.
53. Friedmann H. From colonialism to green capitalism: social movements and emergence of food regimes. *New directions in the sociology of global development*. Emerald Group Publishing Limited; 2005. pp. 227–64.
54. Tilzey M. Food Regimes, Capital, State, and class: Friedmann and mcmichael revisited. *Sociologia Ruralis*. 2019;59(2):230–54.
55. Bless A. The co-optation of regenerative agriculture: revisiting the corporate environmental food regime. *Globalizations*. 2024;1–23.
56. Patel R. Food sovereignty. *J Peasant Stud*. 2009;36(3):663–706.
57. Howard PH. Concentration and power in the food system. *Concentration and Power in the Food System*. 2021;1–232.
58. Baker P, Friel S. Food systems transformations, ultra-processed food markets and the nutrition transition in Asia. *Globalization Health*. 2016;12(1).
59. World Bank. *Population estimates and projections Washington D.C.*: World Bank. 2024.
60. Refinitiv, S&P Compustat. *fundamentals*. 2024. [Available from: <https://www.lseg.com/en/data-analytics/financial-data/company-data/fundamentals-data/standardized-fundamentals/sp-compustat-database>]
61. World Bank. *GDP deflator (United States of America)*. In: World Bank, editor. Online. 2025.
62. Popkin BM. Measuring the nutrition transition and its dynamics. *Public Health Nutr*. 2021;24(2):318–20.
63. Nasdaq. *Dividend history - Chipotle Mexican Grill, Inc.* Online. 2025 [Available from: <https://www.nasdaq.com/market-activity/stocks/cmrg/dividend-history>]
64. Shake Shack. *FAQ Online*. 2025. [Available from: <https://investor.shakeshack.com/resources/faqs/default.aspx#:~:text=Does%20Shake%20Shack%20pay%20a,Noj>]
65. Richardson AA. The effect of franchising on local restaurants in emerging markets: A case study. *Northcentral University*; 2011.
66. Beamish PW, Killing JP, Cooperative Strategies. In: Beamish PW, Killing JP, editors. *European perspectives*. United States: New Lexington.: New Lexington Press. 1997.
67. Zeller Z. They're lovin'it: how China and Japan turned American fast food into global phenomena. *Columbia J Asia*. 2024;3(1):43–70.
68. Wang Y. Localization of mcdonald's and KFC in China. *BCP Bus Manage*. 2023;38:1457–61.
69. Witkowski TH, Ma Y, Zheng D. Cross-cultural influences on brand identity impressions: KFC in China and the united States. *Asia Pac J Mark Logistics*. 2003;15(1/2):74–88.
70. US International Trade Administration. *Saudi Arabia food service sector opportunities for U.S. companies Market Intelligence - US International Trade Administration*. 2023. [Available from: <https://www.trade.gov/market-intelligence/saudi-arabia-food-service-sector-opportunities-us-companies>]
71. Alotaibi A. The role of international Franchises in the economic development of Saudi Arabia. 2019.
72. Rogmans TJ. The emerging markets of the middle east: strategies for entry and growth. *Business Expert*; 2012.
73. Clark SE, Corinna H, MSM E, A. H-KK, and, Wallinga D. Exporting obesity: US farm and trade policy and the transformation of the Mexican consumer food environment. *Int J Occup Environ Health*. 2012;18(1):53–64.
74. Gálvez A. *Eating NAFTA: trade, food policies, and the destruction of Mexico*. Univ of California; 2018.
75. Gómez EJ. *Junk food politics: how beverage and fast food industries are reshaping emerging economies*. JHU; 2023.
76. Ismail A. Domesticating neoliberal foodscapes: an everyday approach to understanding food system transitions in Oaxaca, Mexico. *J Political Ecol*. 2025;32(1).
77. Li MH, Cui L, Lu J. Varieties in state capitalism: Outward FDI strategies of central and local state-owned enterprises from emerging economy countries. *State-Owned Multinationals: Governments in Global Business*. 2018:175–210.
78. Pearson MM, Rithmire M, Tsai K. *The state and capitalism in China*. Cambridge University Press; 2023.
79. CITIC Capital announces an increased investment in McDonald's China, leading the company into a New Era of growth [press release]. CITIC Capital. 2024.
80. Assmann S. *Global recognition and domestic containment: culinary soft power in Japan*. Springer International Publishing. 2017:113–37.
81. Takahashi K, Hashimoto Y. Small grant subsidy application effects on productivity improvement: evidence from Japanese SMEs. *Small Bus Econ*. 2023;60(4):1631–58.
82. Bueno N. Corporate liability for violations of the human right to just conditions of work in extraterritorial operations. *Int J Hum Rights*. 2017;21(5):565–88.
83. Hardy T. Shifting risk and shirking responsibility? The challenge of upholding employment standards regulation within franchise networks. *Australian J Labour Law*. 2019;32(1):62–82.
84. Seeking Alpha. *Yum needs to stop buying back its stock by issuing debt online*. 2019. [Available from: <https://seekingalpha.com/article/4251346-yum-needs-to-stop-buying-back-stock-issuing-debt>]
85. Lawrence B, Pietrafesa C, Kaufmann PJ. *Exploring the growth of multi-unit franchising*. Handbook of research on franchising. Edward Elgar Publishing. 2017:94–115.
86. Tucker E, Labour. *Labour law and capitalist rent seeking: rentier capitalism and labour in historical perspective*. Global Rental Capitalism: Theory and Development (Routledge, forthcoming 2024), Osgoode Legal Studies Research Paper. 2023(4799759).
87. Tucker E, Bartkiw TJ. Franchising and the extraction of surplus value: excavating the legal boundary between franchisees and employees. *J Law Political Econ*. 2023;3(3).
88. Lazonick W, Hopkins M, Jacobson K. McDonald's has to do more than manipulate its stock price. Online: *Havard Business Review*; 2015.
89. Braun B. Exit, control, and politics: structural power and corporate governance under asset manager capitalism. *Politics Soc*. 2022;50(4):630–54.
90. Friel S, Schram A, Frank N, Arthur M, Townsend B, Gajurel H. Financialisation: a 21st century commercial determinant of health equity. *Lancet Public Health*. 2024;9(9):e705–8.
91. Christophers B. *Rentier capitalism: who owns Economy, and who pays for it?* Verso Books; 2020.
92. Kapoor A, Kumar N. Need for franchising laws in India. *Economic Advisory Council to the PM; Institute for Competitiveness*; 2024.
93. Welsh DH, Alon I, Falbe CM. An examination of international retail franchising in emerging markets. *J Small Bus Manage*. 2006;44(1):130–49.
94. Alsin A. McDonald's: burgers, fries and stock buybacks. *Forbes*. 2017.
95. Baertlein L, Yum Brands P, Hut. KFC parent, returns \$US13.5b to shareowners by 2019. *Australian Financial Rev*. 2016.
96. Mazzucato M. *Financialization of the real economy. The value of everything: making and taking in the global economy*. Hachette UK; 2018.
97. Wood B, Sacks G. The influence of share buybacks on ill-health and health inequity: an exploratory analysis using a socio-ecological determinants of health lens. *Globalization Health*. 2023;19(1).
98. Appelbaum E, Batt R. *Private equity at work: when wall street manages main street*. Russell Sage Foundation; 2014.
99. Doctorow C. *Pluralistic* [Internet]. Doctorow C, editor. Online. 2024. [cited 2024]. Available from: <https://doctorow.medium.com/https-pluralistic-net-2024-05-23-spineless-invertebrates-ac092ac1de39>
100. Bitter A. The restaurant brands that PE firm roark capital owns. *Business Insider*. 2024.

101. Roark Capital. Portfolio companies: Franchise / Multi-Unit Online: Roark Capital. 2024 [Available from: https://www.roarkcapital.com/portfolio#s1_Franchise_Multi-Unit]
102. Amado ML. Neoliberalism and labor displacement in panama: contested public space and the disenfranchisement of street vendors. Lexington Books; 2024.
103. Jaffe J, Gertler M. Victual vicissitudes: consumer deskilling and the (gendered) transformation of food systems. *Agric Hum Values*. 2006;23:143–62.
104. Parker C, Johnson H. From food chains to food webs: regulating capitalist production and consumption in the food system. *Annual Rev Law Social Sci*. 2019;15(1):205–25.
105. Lu J. Multiple modernities and multiple proximities: mcdonald's and Kentucky fried chicken in Chinese television commercials. *Int Communication Gaz*. 2010;72(7):619–33.
106. Van Der Laan S, Breeman G, Scherer L. Animal lives affected by meat consumption trends in the G20 countries. *Animals*. 2024;14(11):1662.
107. HLPE. Strengthening urban and peri-urban food systems to achieve food security and nutrition, in the context of urbanization and rural transformation. CFS: Rome: Committee on World Food Security: High Level Panel of Expert;; 2024.
108. Otterbach S, Oskorouchi HR, Rogan M, Qaim M. Using Google data to measure the role of big food and fast food in South africa's obesity epidemic. *World Dev*. 2021;140:105368.
109. Biswas M. Seeds of neocolonialism in development discourse: a study of neoliberal megarethorics of global development and ecofeminist resistance. 2017.
110. Lawrance B, de la Peña C. Local foods Meet global foodways. New York: Routledge; 2012.
111. Yates J, Kadiyala S, Deeney M, Carriedo A, Gillespie S, Heindel JJ, et al. A toxic relationship: ultra-processed foods & plastics. *Globalization Health*. 2024;20(1).
112. Howes K, Shahid M, Jones A, Dunford E, Sacks G, Pettigrew S. FoodSwitch: the state of the fast Food Supply. 2020.
113. Brown H, Kirkman S, Albani V, Goffe L, Akhter N, Hollingsworth B, et al. The impact of school exclusion zone planning guidance on the number and type of food outlets in an english local authority: A longitudinal analysis. *Health Place*. 2021;70:102600.
114. Khan LM, Vaheesan S. Market power and inequality: The antitrust counter-revolution and its discontents. *Harv L & Pol'y Rev*. 2017;11:235.
115. Wood B, Lacy-Nichols J, Sacks G. Taking on the corporate determinants of Ill-Health and health inequity: a scoping review of actions to address excessive corporate power to protect and promote the public's health. *Int J Health Policy Manage*. 2023;12.
116. Hoffman Lent K, Schwartz K. Antitrust enforcement against roll-ups and serial acquisitions. *New York Law Journal*. 2024.
117. Khan LM. It's time to halt roll-up schemes that violate antitrust laws. *Financial Times*. 2023.
118. Elliott JG. Corporate Sustainability due diligence. 2023.
119. Hawkes C, Walton S, Haddad L, Fanzo J. 42 policies and actions to orient food systems towards healthier diets London: centre for food policy. 2020 [Available from: <https://www.foodsystemsdashboard.org/policies-and-actions>]
120. Rizvi A. Guaranteeing a place at the table for everyone: an investigation of food Insecurity, Poverty, and the potential benefits of a guaranteed basic income. Université d'Ottawa| University of Ottawa; 2024.
121. Wood B, McLean A, Sacks G. Health in the age of asset manager capitalism. *Health Promot Int*. 2024;39(5):daae124.
122. Keeble M, Adams J, Sacks G, Vanderlee L, White CM, Hammond D, Burgoine T. Use of online food delivery services to order food prepared Away-From-Home and associated sociodemographic characteristics: A Cross-Sectional, Multi-Country analysis. *Int J Environ Res Public Health*. 2020;17(14):5190.
123. Bennett, R., Sacks, G., Zorbas, C., Huse, O., Wood, B., & Backholer, K. Exploring the commercial determinants of health in the online food delivery sector: a case study of Uber Eats in Australia. *Crit Public Health*. 2025;35(1):2469776.
124. Northcott T, Sievert K, Russell C, Obeid A, Angus D, Parker C. Unhealthy food advertising on social media: policy lessons from the Australian ad observatory. *Health Promot Int*. 2025;40(2).

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Terms and Conditions

Springer Nature journal content, brought to you courtesy of Springer Nature Customer Service Center GmbH (“Springer Nature”).

Springer Nature supports a reasonable amount of sharing of research papers by authors, subscribers and authorised users (“Users”), for small-scale personal, non-commercial use provided that all copyright, trade and service marks and other proprietary notices are maintained. By accessing, sharing, receiving or otherwise using the Springer Nature journal content you agree to these terms of use (“Terms”). For these purposes, Springer Nature considers academic use (by researchers and students) to be non-commercial.

These Terms are supplementary and will apply in addition to any applicable website terms and conditions, a relevant site licence or a personal subscription. These Terms will prevail over any conflict or ambiguity with regards to the relevant terms, a site licence or a personal subscription (to the extent of the conflict or ambiguity only). For Creative Commons-licensed articles, the terms of the Creative Commons license used will apply.

We collect and use personal data to provide access to the Springer Nature journal content. We may also use these personal data internally within ResearchGate and Springer Nature and as agreed share it, in an anonymised way, for purposes of tracking, analysis and reporting. We will not otherwise disclose your personal data outside the ResearchGate or the Springer Nature group of companies unless we have your permission as detailed in the Privacy Policy.

While Users may use the Springer Nature journal content for small scale, personal non-commercial use, it is important to note that Users may not:

1. use such content for the purpose of providing other users with access on a regular or large scale basis or as a means to circumvent access control;
2. use such content where to do so would be considered a criminal or statutory offence in any jurisdiction, or gives rise to civil liability, or is otherwise unlawful;
3. falsely or misleadingly imply or suggest endorsement, approval, sponsorship, or association unless explicitly agreed to by Springer Nature in writing;
4. use bots or other automated methods to access the content or redirect messages
5. override any security feature or exclusionary protocol; or
6. share the content in order to create substitute for Springer Nature products or services or a systematic database of Springer Nature journal content.

In line with the restriction against commercial use, Springer Nature does not permit the creation of a product or service that creates revenue, royalties, rent or income from our content or its inclusion as part of a paid for service or for other commercial gain. Springer Nature journal content cannot be used for inter-library loans and librarians may not upload Springer Nature journal content on a large scale into their, or any other, institutional repository.

These terms of use are reviewed regularly and may be amended at any time. Springer Nature is not obligated to publish any information or content on this website and may remove it or features or functionality at our sole discretion, at any time with or without notice. Springer Nature may revoke this licence to you at any time and remove access to any copies of the Springer Nature journal content which have been saved.

To the fullest extent permitted by law, Springer Nature makes no warranties, representations or guarantees to Users, either express or implied with respect to the Springer nature journal content and all parties disclaim and waive any implied warranties or warranties imposed by law, including merchantability or fitness for any particular purpose.

Please note that these rights do not automatically extend to content, data or other material published by Springer Nature that may be licensed from third parties.

If you would like to use or distribute our Springer Nature journal content to a wider audience or on a regular basis or in any other manner not expressly permitted by these Terms, please contact Springer Nature at

onlineservice@springernature.com